

$$S = - N k_B \log \left[\frac{N}{V} \left(\frac{1}{2 \pi m k_B T} \right)^{3/2} \right] + \frac{3}{2} N k_B$$

$$\frac{\partial f}{\partial t} + \vec{v} \cdot \vec{\nabla}_r f + \vec{a} \cdot \vec{\nabla}_v f = \left(\frac{\partial f}{\partial t} \right)_{coll.}$$

$$\dot{x} = \sigma (y - x)$$

$$\dot{y} = \rho x - y - xz$$

$$\dot{z} = -\beta z + xy$$

$$\frac{\partial f_n}{\partial t} = \{H_n, f_n\} + \sum_{i=1}^n \int d^3r_{n+1} d^3p_{n+1} \frac{\partial U(\vec{r}_i - \vec{r}_{n+1})}{\partial \vec{r}_i} \cdot \frac{\partial f_{n+1}}{\partial \vec{p}_i}$$

$$|BES\rangle = \prod_k (u_k + v_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger) |vac\rangle$$

$$\rho \left(\frac{\partial}{\partial t} + \vec{u} \cdot \vec{\nabla} \right) \vec{u} = \rho \frac{\vec{F}}{m} - \vec{\nabla} P + \eta \nabla^2 \vec{u} + \left(\frac{1}{3} + S \right) \vec{\nabla} (\vec{\nabla} \cdot \vec{u})$$

$$R^{\mu\nu} - \frac{1}{2} g^{\mu\nu} R = \frac{8\pi G}{c} T^{\mu\nu}$$

$$Q_N(V,T) = \frac{1}{N! h^{3N}} \int d^{3N}p d^{3N}r \exp \left(-\beta \sum_i \frac{p_i^2}{2m} - \beta \sum_{i < j} v_{ij} \right)$$

$$\mathcal{L}(x) = \frac{1}{2} \partial^\mu \phi(x) \partial_\mu \phi(x) - \frac{1}{2} m^2 \phi^2(x)$$

$$\mathcal{L}_{QED} = \bar{\psi}(i\not{\partial} - m)\psi + i q \bar{\psi} \gamma^\mu A_\mu \psi - \frac{1}{4} F^{\mu\nu} F_{\mu\nu}$$

$$\rho = \frac{e^{-\beta \mathcal{H}}}{Z}$$

$$\left[-\frac{\hbar^2}{2m} \nabla_i^2 + v_{ion}(\vec{r}_i) + e^2 \sum_{j \neq i} \int d^3r_j \frac{|\phi_j(\vec{r}_j)|^2}{|\vec{r}_i - \vec{r}_j|} \right] \phi_i(\vec{r}_i) - e^2 \sum_{j \neq i} \int d^3r_j \frac{\phi_i^*(\vec{r}_i) \phi_j(\vec{r}_j)}{|\vec{r}_i - \vec{r}_j|} \phi_j(\vec{r}_i) = \varepsilon_i \phi_i(\vec{r}_i)$$

$$\frac{d\vec{P}}{dt} + \frac{d\vec{G}}{dt} = \oint_S (\vec{T}_{em} \cdot \vec{n}) dS$$

$$Z = \int \mathcal{D} \phi(\vec{x}) \; e^{-(\mathcal{H} - \int d^d x \; h(\vec{x}) \phi(\vec{x}))} \mathcal{Z}$$

$$\mathcal{L} = \frac{\epsilon_0}{2} \vec{E} \cdot \vec{E} + \frac{1}{2\mu_0} \vec{B} \cdot \vec{B} + \vec{j} \cdot \vec{A} - eV$$

$$\varepsilon(\vec{q},\omega) = 1 - \frac{e^2}{\epsilon_0 q^2 V} \sum_{\vec{k},\sigma} \frac{f_{\vec{k}+\vec{q}} - f_{\vec{k}}}{\epsilon_{\vec{k}+\vec{q}} - \epsilon_{\vec{k}} + \hbar\omega + i\eta}$$

$$F_{LG} = \int d^3x \left[\frac{1}{2} (\vec{r} \cdot \vec{v})^2 + c \left(\left(\vec{\nabla} - \frac{2ie}{\hbar c} \vec{A} \right) \psi \right)^2 + \frac{1}{2} g |\psi|^4 \right] + \frac{1}{8\pi\mu_0} \int d^3x (\vec{\nabla} \times \vec{A})^2 - \frac{1}{4\pi} \int d^3x \vec{H} \cdot (\vec{\nabla} \times \vec{A})$$

$$\frac{df(r,t)}{dt} = \frac{d}{dr} (r \sigma f(r,t)) + \frac{1}{2} a \frac{d^2}{dr^2} f(r,t)$$

PhD Thesis

$$\Psi(\vec{r},t) = \sqrt{m c(t)} \; e^{iS(\vec{r},t)/\hbar}$$

$$\frac{\partial n}{\partial t} = \nabla^2 n$$

Optical Properties of Two-dimensional Semiconductors: Excitonic and Polaritonic Effects

$$\Gamma_{\mu\nu}^{\lambda} = \frac{1}{2} g^{\lambda\sigma} [g_{\sigma\mu,\nu} + g_{\sigma\nu,\mu} - g_{\mu\nu,\sigma}]$$

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$$S = k_B \log \Omega$$

$$\oint_C f(z) dz = 0$$